

Part A: Conceptual Understanding

1. What are Supervised Learning Algorithms?

Supervised Learning is a type of Machine Learning where the model learns from labeled data. The dataset contains both input values and correct output values. The model uses this data to make predictions on new data.

2. Difference Between Regression and Classification

Regression:

- Predicts numerical values
- Output is continuous
- Example: House Price Prediction

Classification:

- Predicts categories
- Output is discrete
- Example: Spam Detection

3. Explain Simple Linear Regression

Simple Linear Regression is used to find the relationship between one independent variable and one dependent variable. It predicts values using a straight-line equation.

Formula:

$$y = mx + c$$

Where:

y = Predicted value

x = Independent variable

m = Slope

c = Intercept

4. What are the Assumptions of Linear Regression?

1. Linearity – Variables should have a linear relationship.
2. Independence – Errors should be independent.
3. Homoscedasticity – Error variance should remain constant.
4. Normality – Errors should be normally distributed.
5. No Multicollinearity – Features should not be highly correlated.
6. No Significant Outliers – Extreme values should be limited.

5. What is Bias-Variance Trade-Off?

Bias occurs when a model is too simple and misses patterns in data.

Variance occurs when a model learns the training data too closely.

A good model maintains a balance between bias and variance.

6. Explain Overfitting and Underfitting

Overfitting:

The model learns the training data too closely and performs poorly on new data.

Underfitting:

The model is too simple and cannot learn important patterns from the data.